Panel Data: Fixed Effects, Random Effects and Correlated Random Effects

Notes from *From Chance to Decisions*, Unit 11

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Every regression so far has used a cross-section: many units, each observed once. A worker, a household, a district, a firm — one row each.

Unit 8 tried to make such comparisons fair by measuring the other differences between units and holding them constant. It could do that only for what we measured. The farmer's skill, the quality of the soil, the honesty of the district administration, the management culture of the firm — the differences most likely to matter are exactly the ones that never appear in a survey.

Now suppose each unit is observed **more than once**. The same farms, season after season. The same states, year after year. The same households, revisited.

That changes what is possible. A farm observed twice can be compared **with itself**. Whatever about the farm stays the same between the two visits — its soil, its location, its owner's temperament — cancels out of the comparison, whether or not anyone ever measured it.

This unit is about how to use that idea properly. We spend most of our time on one estimator, the **fixed effects estimator**, because understanding what it does — and what it quietly throws away — is the heart of panel data analysis. Two others follow: **random effects**, which is more efficient when its assumption holds, and **correlated random effects**, which sits between them and turns out to be the most useful of the three.

The same unit, observed again

> *Does fertiliser raise yield — or do farms with good soil simply use more of it?*

Suppose we want to know how much fertiliser raises crop yield. A survey of 1,000 farms in a single season records yield and fertiliser use, and the two are strongly positively related.

Is that the effect of fertiliser?

Not necessarily. Farms with better soil get higher yields whatever they do, and they may also find fertiliser more worthwhile, so they use more of it. Soil quality is then driving both numbers. The cross-section cannot separate the two stories, and soil quality is not in the survey to control for.

Recall the thought experiment of Unit 6: the same student, observed twice, differing only in attendance. A cross-section can never supply it, because no unit appears twice. A panel can supply something close to it.

Follow one farm through five seasons. Its soil does not change. If its yield rises in the seasons it applies more fertiliser, the soil cannot be the explanation, because the soil was the same in all five seasons. The comparison is between the farm and itself.

> **KEY IDEA**
> **Panel data** follow the same units over several time periods. Their value is that a unit can be compared with itself, which removes every characteristic that stays fixed over time — measured or not.

What panel data look like

Panel data are usually stored in **long form**: one row for each unit in each period.

To learn how the methods behave we need to know the truth, so we begin with a simulated panel in which we do. This is the same device Unit 3 used for confidence intervals: build a world, run the method, and compare with what we built.

```r
make_panel <- function(N = 200, T = 5, beta = 0.30, rho = 0.6, gamma = 0.20,
                       seed = NULL) {
  if (!is.null(seed)) set.seed(seed)
  a     <- rnorm(N, sd = 0.5)            # soil quality: never observed
  habit <- rnorm(N, sd = 0.5)            # a farm's lasting fertiliser habit
  canal <- rbinom(N, 1, 0.4)             # on a canal: fixed, observed
  d <- data.frame(farm = rep(1:N, each = T), season = rep(1:T, times = N))
  d$canal  <- canal[d$farm]
  d$lfert  <- 1 + rho * a[d$farm] + habit[d$farm] + rnorm(N * T, sd = 0.5)
  d$lyield <- 2 + beta * d$lfert + gamma * d$canal + a[d$farm] +
              rnorm(N * T, sd = 0.3)
  d$soil   <- a[d$farm]                  # kept only so we can check our work
  d
}

farms <- make_panel(seed = 11)
head(farms[, c("farm", "season", "lfert", "lyield", "canal")], 10)
```

```
#>    farm season  lfert lyield canal
#> 1     1      1 1.6244  2.492     1
#> 2     1      2 0.4802  1.724     1
#> 3     1      3 1.3579  2.859     1
#> 4     1      4 1.6182  2.277     1
#> 5     1      5 0.6892  2.452     1
#> 6     2      1 2.1401  3.000     0
#> 7     2      2 1.4556  2.858     0
#> 8     2      3 1.9642  2.603     0
#> 9     2      4 1.7863  2.738     0
#> 10    2      5 1.0471  1.960     0
```

There are 200 farms observed over 5 seasons, so 1000 rows. `lfert` is the log of fertiliser applied and `lyield` the log of yield. A panel in which every unit appears in every period is **balanced**; otherwise it is **unbalanced**.

The truth is built in. Each extra 1% of fertiliser raises yield by **0.30%**: the true β_1 is 0.30. Soil quality `soil` (a_i) also raises yield, and farms with better soil use more fertiliser (`rho = 0.6`). The column `soil` is kept only so that we can check our answers; a real survey would not have it.

The model

Write the outcome for unit i in period t as

$$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + u_{it}, \qquad i = 1, \ldots, N, \quad t = 1, \ldots, T$$

There are two kinds of unobserved influence here, and keeping them apart is the whole subject.

KEY IDEA

- a_i is the **unobserved effect**: everything about unit i that is constant over time and not in the model. It has no t subscript.
- u_{it} is the **idiosyncratic error**: whatever changes from period to period. A good season, a pest outbreak, a delayed delivery of seed.

Their sum, $v_{it} = a_i + u_{it}$, is the **composite error**. Pooled OLS treats v_{it} as a single error term.

Two sources of variation

Any variable in a panel varies in two ways: **between** units (farm 7 uses more fertiliser than farm 8, on average) and **within** a unit over time (farm 7 used more this season than last). The decomposition is exact.

```
demean <- function(v, g) v - ave(v, g)          # subtract each unit's own mean

within_sd  <- function(v, g) sd(demean(v, g))
between_sd <- function(v, g) sd(tapply(v, g, mean))

round(rbind(
  lfert  = c(overall = sd(farms$lfert),
             between = between_sd(farms$lfert, farms$farm),
             within  = within_sd(farms$lfert, farms$farm)),
  lyield = c(overall = sd(farms$lyield),
             between = between_sd(farms$lyield, farms$farm),
             within  = within_sd(farms$lyield, farms$farm))), 3)
```

```
#>        overall between within
#> lfert    0.763   0.617  0.450
#> lyield   0.683   0.616  0.298
```

Fertiliser use differs a good deal between farms and less within a farm over time. This matters because, as we are about to see, the fixed effects estimator uses **only the within column**.

Why pooling fails

> *What goes wrong if we ignore the panel structure and run OLS on all 1000 rows?*

Pooled OLS stacks every observation and regresses y on x as if each row were an independent draw.

```
pooled <- lm(lyield ~ lfert, data = farms)
round(coef(summary(pooled)), 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   1.8490     0.0286   64.59        0
#> lfert         0.5248     0.0230   22.84        0
```

The estimate is 0.525, against a truth of 0.30. Two different things have gone wrong.

Problem 1: bias

The soil quality a_i sits inside the composite error v_{it}. If it is correlated with fertiliser use, pooled OLS violates the zero conditional mean assumption exactly as in Unit 8: the omitted variable bias formula applies, with a_i as the omitted variable.

$$\operatorname{plim} \hat{\beta}_1^{\text{pooled}} = \beta_1 + \frac{\operatorname{Cov}(x_{it}, a_i)}{\operatorname{Var}(x_{it})}$$

We can check this against the simulation, which tells us every ingredient.

```
cov_xa <- 0.6 * 0.5^2                    # rho times Var(a)
var_x  <- 0.6^2 * 0.5^2 + 0.5^2 + 0.5^2 # soil + habit + season-to-season noise
round(c(truth = 0.30, bias = cov_xa / var_x,
        predicted = 0.30 + cov_xa / var_x,
        pooled_estimate = unname(coef(pooled)[2])), 3)
```

```
#>           truth            bias       predicted pooled_estimate
#>           0.300           0.254           0.554           0.525
```

The estimate is far above the truth, and close to what the formula predicts (this is one sample; the average over many panels, in the Monte Carlo section, is 0.555). It would be mistaken to call this a small-sample accident: a bigger panel would only estimate the wrong number more precisely.

Problem 2: standard errors

The second problem is separate, and it would remain even if there were no bias. The same a_i appears in all five of a farm's composite errors, so those errors are **correlated with one another**. Pooled OLS assumes they are independent.

```
farms$res_pooled <- resid(pooled)
wide <- reshape(farms[, c("farm", "season", "res_pooled")],
                idvar = "farm", timevar = "season", direction = "wide")
round(cor(wide[, -1])[1:3, 1:3], 2)
```

```
#>              res_pooled.1 res_pooled.2 res_pooled.3
#> res_pooled.1         1.00         0.70         0.71
#> res_pooled.2         0.70         1.00         0.66
#> res_pooled.3         0.71         0.66         1.00
```

Residuals from the same farm in different seasons correlate at about 0.7. Treating 1000 rows as 1000 independent pieces of information overstates how much we know, and the usual standard errors are too small — the same lesson as Unit 7, for a different reason.

The remedy is the **cluster-robust standard error**, which allows arbitrary correlation among the observations of one unit and treats the unit, not the row, as the independent draw.

```
cluster_se <- function(m, cluster) {
  X <- model.matrix(m); u <- residuals(m); G <- length(unique(cluster))
  bread <- solve(crossprod(X))
  meat  <- crossprod(rowsum(X * u, cluster))   # sum scores within each unit
  sqrt(diag(G / (G - 1) * bread %*% meat %*% bread))
}

round(c(usual = coef(summary(pooled))[2, 2],
        clustered = cluster_se(pooled, farms$farm)[2]), 4)
```

```
#>           usual clustered.lfert
#>           0.023           0.035
```

The clustered standard error is about 50% larger. Cluster at the level of the unit that is observed repeatedly, and use these standard errors for every estimator in this unit.

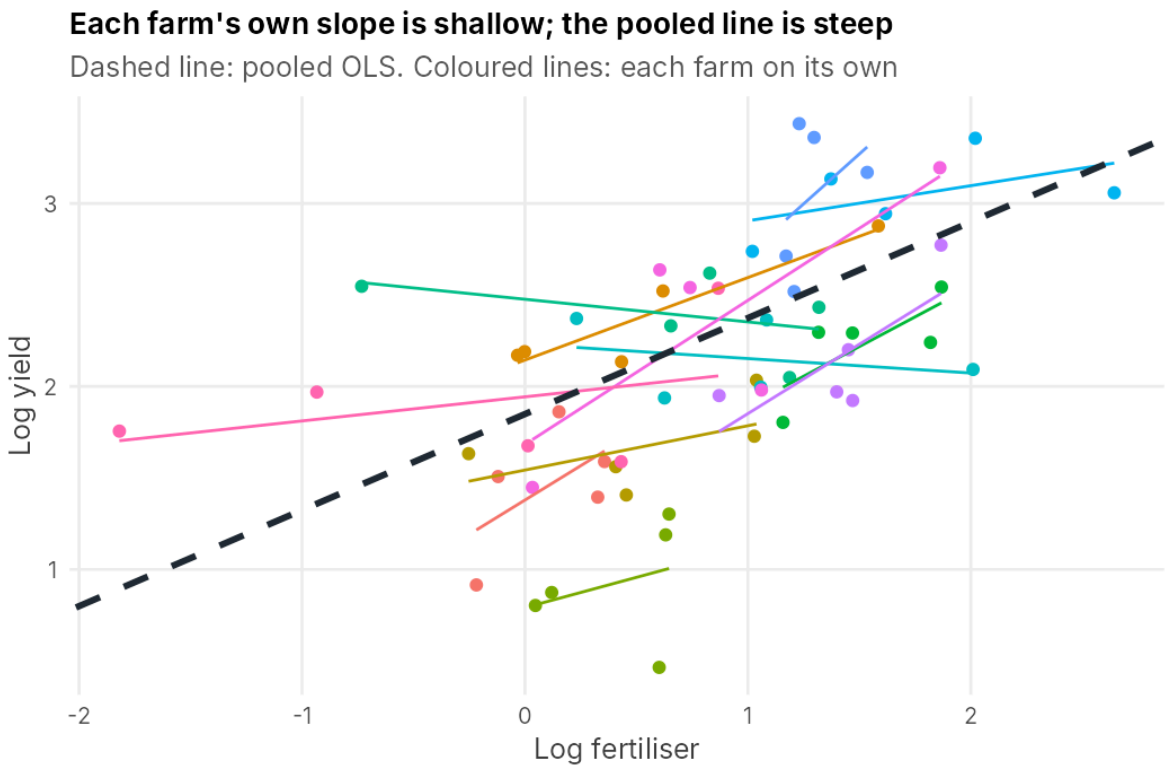
Clustering repairs only problem 2. It does nothing for problem 1: an estimate that is centred on the wrong number is no better for having an honest standard error.

Seeing it

Plot twelve farms. Each farm has its own cloud of five points, and its own trend within the cloud.

```
set.seed(21)
pick <- sample(unique(farms$farm), 12)
sub  <- farms[farms$farm %in% pick, ]

ggplot(sub, aes(lfert, lyield, colour = factor(farm), group = farm)) +
  geom_point(size = 2, show.legend = FALSE) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.6, show.legend = FALSE) +
  geom_abline(intercept = coef(pooled)[1], slope = coef(pooled)[2],
              colour = book_palette$ink, linewidth = 1.3, linetype = "dashed") +
  labs(x = "Log fertiliser", y = "Log yield",
       title = "Each farm's own slope is shallow; the pooled line is steep",
       subtitle = "Dashed line: pooled OLS. Coloured lines: each farm on its own") +
  theme_book()
```



plot of chunk pd-lines

Within any one farm the relationship is flat-ish — about 0.30. Across farms the clouds are stacked: farms to the right sit higher, because they have better soil. The dashed pooled line mixes the two, so it is far steeper than any single farm's own line. The fixed effects estimator is an average of the farms' *own* lines.

The fixed effects estimator

> *What exactly does the fixed effects estimator do — and what does it throw away?*

The within transformation

Start with the model for one unit, and **average it over time**:

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_i + a_i + \bar{u}_i, \qquad \text{where } \bar{y}_i = \frac{1}{T}\sum_{t=1}^{T} y_{it}$$

Because a_i does not change with t, it appears in the averaged equation exactly as it appeared in the original. Now **subtract** the averaged equation from the original, period by period:

THE MATHS

$$
\begin{aligned}
y_{it} - \bar{y}_i &= \beta_1 (x_{it} - \bar{x}_i) + (a_i - a_i) + (u_{it} - \bar{u}_i) \\
\ddot{y}_{it} &= \beta_1 \ddot{x}_{it} + \ddot{u}_{it}
\end{aligned}
$$

where a double dot marks a variable **minus its unit's own time average**. The intercept β_0 cancels too, and so does a_i.

This is the **fixed effects transformation**, also called the **within transformation**. The unobserved effect has vanished from the equation. Whatever a_i is, and however it is related to x, it cannot bias an estimate made from this equation because it is no longer there.

The **fixed effects (within) estimator** is OLS on the transformed variables:

$$
\hat{\beta}_1^{FE} = \frac{\sum_i \sum_t \ddot{x}_{it} \ddot{y}_{it}}{\sum_i \sum_t \ddot{x}_{it}^2}
$$

Nothing about it is exotic. It is the slope formula of Unit 7, applied to deviations from each unit's own mean instead of deviations from the overall mean.

```
        farms$y_dd <- demean(farms$lyield, farms$farm)
        farms$x_dd <- demean(farms$lfert,  farms$farm)

        fe_by_hand <- sum(farms$x_dd * farms$y_dd) / sum(farms$x_dd^2)
        c(fixed_effects = fe_by_hand, pooled = unname(coef(pooled)[2]), truth = 0.30)
```

```
#> fixed_effects        pooled         truth
#>        0.2938        0.5248        0.3000
```

The fixed effects estimate is 0.294, close to the truth, where pooled OLS gave 0.525.

INTUITION

A farm with good soil has high y in all five seasons and high x in all five. Subtracting the farm's own averages shifts the whole cloud of that farm's points to the origin, wherever it started. Good soil and bad soil are now the same cloud. All that survives is how the farm moves around *its own centre*: a season with more fertiliser than usual against a season with less.

The comparison is the one we wanted: each farm against itself.

The identifying assumption is **strict exogeneity**: given a_i, the idiosyncratic error in *every* period is uncorrelated with the explanatory variable in *every* period. What the estimator does **not** require is that a_i be uncorrelated with x. It permits arbitrary correlation between the unobserved effect and the explanatory variables, in every period. That is its whole advantage.

The same number, three ways

The fixed effects estimate can be reached by several routes. They agree exactly.

Route 1: the demeaned regression is the one above.

Route 2: dummy variables. Treat each farm's a_i as a parameter to be estimated — a separate intercept for each farm — by including a dummy for every farm, exactly as in Unit 10.

$$y_{it} = \beta_1 x_{it} + \underbrace{\alpha_1 D1_i + \alpha_2 D2_i + \cdots + \alpha_N DN_i}_{\text{one intercept per farm}} + u_{it}$$

```
        lsdv <- lm(lyield ~ lfert + factor(farm), data = farms)

        c(demeaned = fe_by_hand, dummies = unname(coef(lsdv)["lfert"]))
```

```
#> demeaned  dummies
#>   0.2938   0.2938
```

```
        length(coef(lsdv))
```

```
#> [1] 201
```

The coefficient on `lfert` is identical to every digit. It is the **dummy variable regression**, or LSDV (least squares dummy variables). With 200 farms the model has 201 parameters. That is why nobody runs it in a large panel: the demeaning route gives the same slope without ever estimating the 200 intercepts.

The agreement is Frisch–Waugh–Lovell (Unit 8) at work. Including a dummy for every farm and then asking for the effect of x is the same as first removing from x and y the part explained by the farm dummies — and what is left after regressing a variable on a full set of unit dummies is precisely its deviation from the unit mean.

Route 3: first differences. Subtract the previous period instead of the unit mean:

$$\Delta y_{it} = y_{it} - y_{i,t-1} \quad \Rightarrow \quad \Delta y_{it} = \beta_1 \Delta x_{it} + \Delta u_{it}$$

a_i cancels here too. With two periods this is the same estimator again. The next sections return to when the two differ.

> **WATCH OUT**
> All three routes remove a_i by comparing a unit only with itself, so all three are **blind to anything constant over time** and all three use only within-unit variation. Their differences are in efficiency and in what they assume about u_{it}, not in what they can see.

The degrees of freedom

The demeaned regression is an ordinary OLS regression on NT rows with one slope, so one might expect $NT - 1$ degrees of freedom, and a standard error computed accordingly. That is wrong.

Within each farm, the demeaned errors $\ddot{u}_{it}$ must add up to zero, because they are deviations from the farm's own average. They are therefore not T free numbers but $T - 1$. Each of the N units costs one degree of freedom, and the correct count is

$$\text{df} = NT - N - k = N(T - 1) - k$$

where k is the number of slopes. An `lm()` on demeaned data does not know this and uses $NT - k$.

```
        n <- nrow(farms); N <- 200; k <- 1
        m_dd <- lm(y_dd ~ x_dd - 1, data = farms)

        s2_naive     <- sum(resid(m_dd)^2) / (n - k)
        s2_corrected <- sum(resid(m_dd)^2) / (n - N - k)

        round(c(
          se_from_lm_on_demeaned = coef(summary(m_dd))[1, 2],
          se_correct_df          = sqrt(s2_corrected / sum(farms$x_dd^2)),
          se_from_dummy_regression = coef(summary(lsdv))["lfert", 2]), 5)
```

```
#>   se_from_lm_on_demeaned            se_correct_df se_from_dummy_regression
#>                  0.01883                  0.02105                  0.02105
```

The dummy-variable regression gets the degrees of freedom right automatically, since it knows it estimated 200 intercepts. A package's fixed effects routine corrects for it as well. The correction matters most when T is small: here it raises the standard error by about 12%, and with $T = 2$ the naive standard error would be too small by a factor of $\sqrt{2}$.

The fit

Two R^2 figures can be reported, and they answer different questions.

```
        r2_within <- summary(m_dd)$r.squared
        r2_lsdv   <- summary(lsdv)$r.squared
        round(c(within = r2_within, with_dummies = r2_lsdv), 3)
```

```
#>       within with_dummies
#>        0.196        0.847
```

The **within** R^2 is the share of the *time variation* in y explained by the time variation in x — the fit of the demeaned equation. The R^2 of the dummy regression is far higher, because 200 farm dummies explain most of the variation between farms. That high number is not an achievement. It says that farms differ a lot, which we knew, and tells us nothing about how well fertiliser explains yield.

The farm intercepts

The unobserved effects themselves can be recovered, because a_i is the part of the farm's average yield that its average fertiliser does not explain:

$$\hat{\alpha}_i = \bar{y}_i - \hat{\beta}_1^{FE}\,\bar{x}_i$$

(These are the dummy coefficients of the LSDV regression. Packages that report an overall "intercept" after fixed effects usually report the average of the $\hat{\alpha}_i$.)

```
        ybar <- tapply(farms$lyield, farms$farm, mean)
        xbar <- tapply(farms$lfert,  farms$farm, mean)
        ahat <- ybar - fe_by_hand * xbar

        truth <- tapply(2 + 0.20 * farms$canal + farms$soil, farms$farm, mean)

        round(c(correlation_with_truth = cor(ahat, truth)), 3)
```

```
#> correlation_with_truth
#>                  0.969
```

The estimated intercepts track the true ones closely, but only closely. Each is based on just $T = 5$ observations, and unlike $\hat{\beta}_1$ they do **not** become reliable as we add farms: a new farm brings a new a_i to be estimated, so no information about any one a_i accumulates. Only a longer T would help. That is why we treat a_i as something to be removed, not as something to be estimated.

What the transformation throws away

Subtracting a unit's average removes a_i, but it removes **anything else that is constant over time** with it. `canal` — whether the farm lies on a canal — is observed, but it never changes for a given farm, so

$$\ddot{canal}_{it} = canal_i - canal_i = 0 \quad \text{for every } i \text{ and } t$$

```
        farms$canal_dd <- demean(farms$canal, farms$farm)
        range(farms$canal_dd)
```

```
#> [1] 0 0
```

```
        coef(lm(y_dd ~ x_dd + canal_dd - 1, data = farms))
```

```
#>     x_dd canal_dd
#>   0.2938       NA
```

The coefficient on `canal` is `NA`: R reports that the variable is a column of zeros and cannot be estimated. In the dummy regression the same thing is visible as perfect collinearity, because `canal` equals a combination of farm dummies.

> **WATCH OUT**
>
> **The fixed effects estimator cannot estimate the effect of any variable that is constant over time.** Gender, caste, a farm's location, years of schooling in a panel of adults, a state's colonial history, distance to a river: all are swept away along with a_i. This is the price of the estimator's main virtue. It cannot be paid back by a larger sample.

Two partial escapes are worth knowing.

First, a time-constant variable can be **interacted with year dummies**. Years of schooling cannot enter, but "schooling × 2019" and "schooling × 2020" can, and their coefficients show how the return to schooling *in each year differs from the return in the base year*. Neither the base-year return nor any year's return can be recovered, only the changes.

Second, the **correlated random effects** approach later in this unit can estimate a coefficient on a time-constant variable while still allowing a_i to be correlated with the time-varying regressors — under a stronger assumption about the time-constant variable.

A related trap: if you include a full set of year dummies, you cannot estimate the effect of a variable whose change over time is the same for everyone. Experience in a panel of workers who are all employed in every year rises by exactly one every year. Its within variation is identical to a time trend, and a time trend is exactly what year dummies capture.

```
farms$experience <- farms$season + 5      # every farm gains one season a year

with_years <- lm(lyield ~ lfert + factor(season) + experience + factor(farm),
                 data = farms)
coef(with_years)["experience"]
```

```
#> experience
#>         NA
```

Again `NA`. The farm effect absorbs how much experience a farm began with; after that, experience cannot be told apart from the season dummies.

Fixed effects or first differences?

Both estimators remove a_i. When do they differ?

With two periods they are the same estimator, provided each is run properly: the first-difference regression needs an intercept (the change in the average outcome between the two periods), which in the fixed effects version appears as a dummy for the second period.

```r
two <- subset(farms, season <= 2)
two <- two[order(two$farm, two$season), ]

dy <- two$lyield[two$season == 2] - two$lyield[two$season == 1]
dx <- two$lfert [two$season == 2] - two$lfert [two$season == 1]

fd_t2 <- coef(lm(dy ~ dx))["dx"]
fe_t2 <- coef(lm(lyield ~ lfert + factor(season) + factor(farm),
                 data = two))["lfert"]
round(c(first_difference = unname(fd_t2), fixed_effects = unname(fe_t2)), 6)
```

```
#> first_difference    fixed_effects
#>           0.3553           0.3553
```

With three or more periods they differ, although both are consistent as N grows. The choice turns on the idiosyncratic errors u_{it}.

```r
farms <- farms[order(farms$farm, farms$season), ]
first_diff <- function(v, g) ave(v, g, FUN = function(z) c(NA, diff(z)))
farms$dy <- first_diff(farms$lyield, farms$farm)
farms$dx <- first_diff(farms$lfert,  farms$farm)

fd_t5 <- coef(lm(dy ~ dx - 1, data = farms))["dx"]
round(c(first_difference = unname(fd_t5), fixed_effects = fe_by_hand), 4)
```

```
#> first_difference    fixed_effects
#>           0.2909           0.2938
```

- If the u_{it} are **serially uncorrelated** — a bad season now says nothing about the next — fixed effects is the more efficient estimator, and its standard errors are the ones to believe. This is the usual maintained assumption.
- If the u_{it} are **highly persistent**, so that shocks accumulate like a random walk, then the *differences* Δu_{it} are uncorrelated and first differencing is better.
- Nothing about the demeaned residuals tells you which holds, because they are not the u_{it}. A test is available in the differenced equation.

When the two estimators give very different answers it is worth reporting both and asking why.

Fixed effects and measurement error

The fixed effects estimator pays a price for working only with within variation. If x is measured with error, the damage depends on how much of the true variation survives. Fertiliser use is recorded from memory; suppose each recorded value carries an error with standard deviation 0.4.

```r
set.seed(3)
one_rep <- function() {
  d <- make_panel()
  d$lfert_noisy <- d$lfert + rnorm(nrow(d), sd = 0.4)
  fe_slope <- function(x) {
    xd <- demean(x, d$farm); yd <- demean(d$lyield, d$farm)
    sum(xd * yd) / sum(xd^2)
  }
  c(clean = fe_slope(d$lfert), noisy = fe_slope(d$lfert_noisy))
}
rowMeans(replicate(300, one_rep()))
```

```
#>  clean  noisy
#> 0.2983 0.1823
```

A noise standard deviation of 0.4 is a modest part of the *overall* variation in fertiliser (standard deviation about 0.8). But a large part of that overall variation is *between* farms, which fixed effects discards. What remains — the within variation — is smaller, so the same noise is a larger share of it and attenuates the slope more. The estimate falls from 0.30 to about 0.18.

> **WATCH OUT**
> Removing the unobserved effect by differencing out the between variation makes the problem of **classical measurement error worse**, not better. Panel methods that remove bias from one source can amplify it from another. If x is noisy and mostly persistent, a fixed effects estimate may be biased towards zero.

Unbalanced panels and attrition

Real panels often have gaps: some units are missing in some periods. Fixed effects handles this without difficulty. Each unit's mean is taken over **its own** periods, one degree of freedom is lost per unit, and the rest is unchanged.

```r
unbal <- farms[!(farms$farm %in% 1:10 & farms$season > 1), ]   # 10 farms seen once
table(Periods_observed = table(unbal$farm))
```

```
#> Periods_observed
#>   1   5
#>  10 190
```

A unit observed only **once** plays no role at all. Its demeaned x and y are exactly zero. In this panel ten farms contribute nothing to the fixed effects estimate, although they remain in the data.

The harder question is *why* the panel is unbalanced. Suppose farms disappear from the survey.

- If a farm drops out for reasons related to a_i — farms on poor soil are more likely to be lost to follow-up — fixed effects is **still fine**, because a_i is removed from the

estimating equation anyway. This is a real advantage over methods that need a_i to be uncorrelated with everything.
- If the farm drops out for reasons related to the **idiosyncratic error** u_{it} — farms are lost precisely because a bad season ruined them — the remaining sample is selected on the outcome, and the estimates are biased.

Inference

Fixed effects standard errors assume u_{it} is homoskedastic and serially uncorrelated. Neither is safe to assume, and both are easy to relax by clustering on the unit, exactly as for pooled OLS.

```
        c(classical = coef(summary(lsdv))["lfert", 2],
          clustered = cluster_se(lsdv, farms$farm)["lfert"])
```

```
#>       classical clustered.lfert
#>         0.02105         0.02197
```

Cluster whenever the unit is observed repeatedly. With few units — say, fewer than about twenty states — cluster-robust standard errors themselves become unreliable, and a long T with small N raises further difficulties not discussed here.

What to remember about fixed effects

KEY IDEA

- The estimator **demeans** each unit's data and runs OLS: the unobserved effect a_i and every other time-constant variable disappear.
- It permits **any** correlation between a_i and the explanatory variables. It needs *strict exogeneity* of the time-varying errors.
- It is identical to OLS with a dummy for each unit; the demeaned version just avoids estimating the dummies. Degrees of freedom are $N(T-1)-k$.
- It uses **only within variation**. Units that never vary, and variables that never vary, contribute nothing.
- With $T = 2$ it equals first differencing; with $T \geq 3$ the two differ, and fixed effects is preferred when u_{it} is not highly persistent.
- It is **not** a cure for omitted variables that *change* over time, and it can aggravate measurement error.

The between estimator

For completeness, the opposite operation: regress unit **averages** on unit averages and use only between variation.

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_i + (a_i + \bar{u}_i)$$

```
        unit <- data.frame(ybar = as.numeric(ybar), xbar = as.numeric(xbar))
        between <- coef(lm(ybar ~ xbar, data = unit))["xbar"]
        round(c(between = unname(between), within = fe_by_hand, truth = 0.30), 3)
```

```
#> between  within   truth
#>   0.648   0.294   0.300
```

The between estimate is far from the truth, because a_i is correlated with $\bar{x}_i$. If a_i is related to x, the between estimator is biased. It is mainly useful as a building block, as the next sections show.

Random effects

> *What if we are willing to assume that the unobserved effect is not correlated with the explanatory variables?*

Fixed effects pays for its robustness by discarding the between variation. If a_i were uncorrelated with x_{it} — if soil quality had nothing to do with how much fertiliser a farm uses — pooled OLS would not be biased, and the between variation would be perfectly good information about β_1. Discarding it would waste data.

The **random effects** model makes that assumption. It treats a_i as a random draw from a population, independent of the regressors:

$$\operatorname{Cov}(x_{it}, a_i) = 0 \quad \text{for all } t$$

Under this assumption pooled OLS is consistent — but still inefficient and with wrong standard errors, since the composite error is serially correlated:

$$\operatorname{Var}(v_{it}) = \sigma_a^2 + \sigma_u^2, \qquad \operatorname{Corr}(v_{it}, v_{is}) = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_u^2} \quad (t \neq s)$$

The efficient way to estimate a regression with an error of this known structure is **generalised least squares**. For this error structure GLS has a particularly simple form: it is OLS after subtracting only *part* of each unit's mean.

> **THE MATHS**
>
> Define the **quasi-demeaned** variables
>
> $$y_{it} - \theta \bar{y}_i, \qquad x_{it} - \theta \bar{x}_i, \qquad \theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2}}$$
>
> Running OLS on these variables, including an intercept (which is multiplied by $1 - \theta$), gives the random effects estimator.

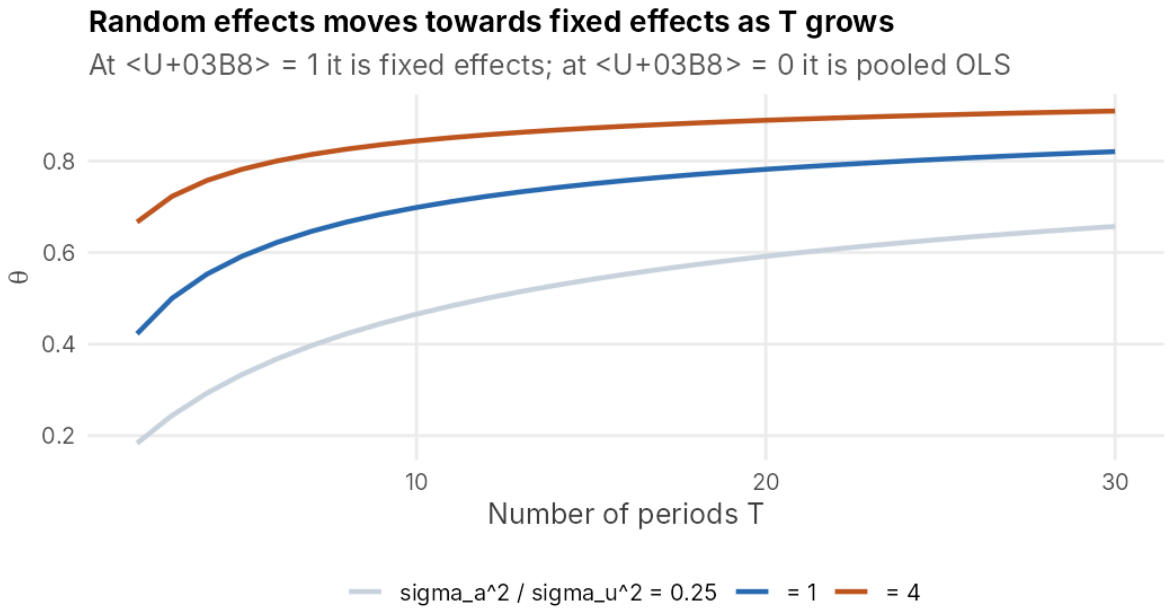
Look at the two extremes of θ.

- $\theta = 0$: nothing is subtracted, and random effects is **pooled OLS**. This happens when $\sigma_a^2 = 0$: there is no unobserved effect to worry about.
- $\theta = 1$: the whole unit mean is subtracted, and random effects is the **fixed effects** estimator. This happens as T or σ_a^2 gets large.

Random effects is therefore a weighted compromise between the within estimator and the between estimator. Where is it on the scale?

```
th <- expand.grid(T = 2:30, ratio = c(0.25, 1, 4))
th$theta <- 1 - sqrt(1 / (1 + th$T * th$ratio))
th$ratio <- factor(th$ratio, labels = c("sigma_a^2 / sigma_u^2 = 0.25",
                                        "= 1", "= 4"))

ggplot(th, aes(T, theta, colour = ratio)) +
  geom_line(linewidth = 1) +
  scale_colour_manual(values = c(book_palette$soft, book_palette$accent,
                                 book_palette$reject), name = NULL) +
  labs(x = "Number of periods T", y = expression(theta),
       title = "Random effects moves towards fixed effects as T grows",
       subtitle = "At θ = 1 it is fixed effects; at θ = 0 it is pooled OLS") +
  theme_book() + theme(legend.position = "bottom")
```



plot of chunk pd-theta

The more periods we have, and the bigger the unobserved effect relative to the idiosyncratic noise, the more random effects behaves like fixed effects.

In practice σ_a^2 and σ_u^2 must themselves be estimated. A simple approach (Swamy–Arora) uses the residual variance from the within regression for σ_u^2, and the between regression to back out σ_a^2.

```r
# Fixed and random effects for one regressor, balanced panel.
fe_re <- function(d, id, y, x) {
  g <- d[[id]]; n <- nrow(d); N <- length(unique(g)); T <- n / N
  yb <- ave(d[[y]], g); xb <- ave(d[[x]], g)
  yd <- d[[y]] - yb;    xd <- d[[x]] - xb

  b_fe  <- sum(xd * yd) / sum(xd^2)
  s2u   <- sum((yd - b_fe * xd)^2) / (n - N - 1)
  se_fe <- sqrt(s2u / sum(xd^2))

  btw   <- lm(yb ~ xb, data = data.frame(yb, xb)[!duplicated(g), ])
  s2a   <- max(sum(resid(btw)^2) / (N - 2) - s2u / T, 0)
  theta <- 1 - sqrt(s2u / (s2u + T * s2a))

  yq <- d[[y]] - theta * yb; xq <- d[[x]] - theta * xb
  re <- lm(yq ~ I(rep(1 - theta, n)) + xq - 1)
  c(b_fe = b_fe, se_fe = se_fe,
    b_re = unname(coef(re)[2]), se_re = coef(summary(re))[2, 2],
    theta = theta, sigma2_a = s2a, sigma2_u = s2u)
}

est <- fe_re(farms, "farm", "lyield", "lfert")
round(est, 4)
```

```
#>     b_fe    se_fe     b_re    se_re    theta sigma2_a sigma2_u
#>   0.2938   0.0211   0.3406   0.0200   0.7151   0.2027   0.0895
```

The estimated variance components are reasonably close to the truth ($\sigma_a^2 = 0.25$, $\sigma_u^2 = 0.09$), and `theta` is 0.72: random effects subtracts about 72% of each farm's mean.

Look at the slope. The random effects estimate is 0.341: higher than the fixed effects estimate of 0.294, and higher than the truth of 0.30. It is not as wrong as pooled OLS, because it removes part of the farm effect, but it is still biased — because soil is correlated with fertiliser, and random effects assumes it is not.

> **WATCH OUT**
> **Random effects is more efficient than fixed effects only if its assumption is true.** If $\text{Cov}(x_{it}, a_i) \neq 0$ it is inconsistent, and a large sample only makes the wrong answer more precise. Fixed effects makes no such assumption.

Fixed or random? The Hausman test

> *Can the data tell us whether the random effects assumption holds?*

Compare the two estimators. If a_i is uncorrelated with x, both are consistent and should give similar answers. If it is correlated, fixed effects is still consistent and random effects is not, and they should differ.

This is the logic of the **Hausman test**. Under H_0 (random effects valid) the two estimators estimate the same number, and the random effects one is efficient:

$$-- \quad --$$

$$H = \frac{(\hat{\beta}^{FE} - \hat{\beta}^{RE})^2}{\widehat{\mathrm{Var}}(\hat{\beta}^{FE}) - \widehat{\mathrm{Var}}(\hat{\beta}^{RE})} \sim \chi^2_1 \quad \text{(for one slope)}$$

A large H rejects H_0: the estimators disagree by more than sampling noise would allow, and fixed effects should be preferred.

```
hausman <- function(e) {
  h <- (e["b_fe"] - e["b_re"])^2 / (e["se_fe"]^2 - e["se_re"]^2)
  c(H = unname(h), p_value = unname(1 - pchisq(h, 1)))
}
round(hausman(est), 5)
```

```
#>       H p_value
#>   49.15    0.00
```

The test rejects emphatically here, as it should, since we built the data to violate the random effects assumption.

WATCH OUT

The Hausman test has three limitations.

1. The variance formula assumes random effects is efficient, which requires the errors to be homoskedastic and serially uncorrelated apart from a_i. With clustering, the denominator can come out **negative**, and the test is invalid.
2. It compares slopes only. It says nothing about whether the fixed effects estimate is *right* — only that the two differ.
3. A failure to reject is not proof that random effects is valid; in small panels the test has little power.

The next section gives a version of the test that does not rely on any of these.

Correlated random effects

> *Is there a way to keep the fixed effects estimate, test the random effects assumption, and still estimate the effect of* `canal`*?*

The **correlated random effects** (CRE) approach, due to Mundlak and Chamberlain, makes the same reasoning that underlies fixed effects explicit.

The random effects problem is that a_i may be related to x_{it}. So model that relationship directly. Suppose a_i depends on the farm's *average* fertiliser use:

$$a_i = \psi \bar{x}_i + c_i, \qquad \mathrm{Cov}(x_{it}, c_i) = 0$$

Here c_i is whatever is left of the unobserved effect after the part related to $\bar{x}_i$ has been taken out. It **is** uncorrelated with x, by construction. Substitute into the model:

$$y_{it} = \beta_0 + \beta_1 x_{it} + \psi \bar{x}_i + c_i + u_{it}$$

This is a random effects model with one extra regressor: the **unit mean of** x. The problem that made random effects biased has been moved into an explicit, estimated term.

```
farms$xbar <- ave(farms$lfert, farms$farm)

cre <- lm(lyield ~ lfert + xbar, data = farms)

round(rbind(
  coef(summary(cre))[c("lfert", "xbar"), 1:2],
  fixed_effects = c(fe_by_hand, NA)), 4)
```

```
#>               Estimate Std. Error
#> lfert           0.2938     0.0379
#> xbar            0.3540     0.0470
#> fixed_effects   0.2938         NA
```

Three results follow from adding $\bar{x}_i$.

KEY IDEA

1. **The coefficient on x_{it} equals the fixed effects estimate** — here 0.2938, to every digit. Controlling for the unit's average makes the remaining variation in x the *within* variation, so the coefficient measures the same thing as the within estimator.
2. **The coefficient on $\bar{x}_i$ is the difference between the between and the within effects**: $\hat{\psi} = \hat{\beta}^{B} - \hat{\beta}^{W}$. A farm using more fertiliser *on average* has a higher yield for reasons that go beyond what extra fertiliser does within a farm.
3. **Testing $\psi = 0$ is a test of the random effects assumption.** If $\psi = 0$, the between and within effects coincide and random effects is valid. It is the Hausman test in regression form.

Checking point 2 against the between estimate computed earlier:

```
round(c(psi_hat = unname(coef(cre)["xbar"]),
        between_minus_within = unname(between) - fe_by_hand), 4)
```

```
#>              psi_hat between_minus_within
#>                0.354                0.354
```

Point 3 has a considerable practical advantage over the Hausman test: the regression can use **cluster-robust standard errors**, so the test remains valid when the errors are heteroskedastic or serially correlated.

```
se_cre <- cluster_se(cre, farms$farm)
t_psi  <- coef(cre)["xbar"] / se_cre["xbar"]
round(c(psi_hat = unname(coef(cre)["xbar"]),
        clustered_se = unname(se_cre["xbar"]),
        t_statistic = unname(t_psi)), 3)
```

```
#>      psi_hat clustered_se  t_statistic
#>        0.354        0.055        6.432
```

t is very large, so $\psi \neq 0$ and the random effects assumption is rejected — the same message as the Hausman test, now in a form that survives clustering.

Recovering time-constant variables

The key practical gain: once $\bar{x}_i$ is in the equation, time-constant variables such as `canal` can be added, and their coefficients **are** estimable.

```
cre2 <- lm(lyield ~ lfert + canal + xbar, data = farms)
round(cbind(estimate = coef(cre2),
            clustered_se = cluster_se(cre2, farms$farm))[2:4, ], 4)
```

```
#>       estimate clustered_se
#> lfert   0.2938       0.0220
#> canal   0.2644       0.0642
#> xbar    0.3486       0.0541
```

The estimated effect of `canal` is 0.264 (standard error about 0.06), against a true value of 0.20, with the fixed effects estimator unable to produce any number at all. The slope on `lfert` is unchanged.

WATCH OUT

This is not free. The coefficient on `canal` is credible only if `canal` is uncorrelated with c_i, the part of the unobserved effect left over after $\bar{x}_i$ is accounted for. In the simulation we made it so — `canal` is independent of soil. In a real panel that is a substantive assumption about the time-constant variable, and it is stronger than anything fixed effects needed. The time-varying coefficient $\hat{\beta}_1$ does **not** depend on it.

GOING DEEPER

With several time-varying regressors, include the unit mean of **each**: $\bar{x}_{1i}, \bar{x}_{2i}, \ldots$. With an **unbalanced** panel the means are taken over each unit's own available periods.

Chamberlain's more general version replaces the single mean by the entire history, $x_{i1}, \ldots, x_{iT}$, which allows a_i to depend on the *pattern* of x over time rather than just its average. Mundlak's version is much easier to use and usually all that is needed.

Adding the means to a pooled OLS regression with clustered standard errors gives the same coefficients as above, in a balanced panel; it is the random-effects weighting that would, in general, differ.

All four on trial

> *Do these four estimators really behave as claimed?*

We have one sample of 200 farms. To see how each estimator behaves in general, simulate many panels and record each estimate. We do it twice: once with soil correlated with fertiliser (as before) and once with `rho = 0`, so that the random effects assumption is true.

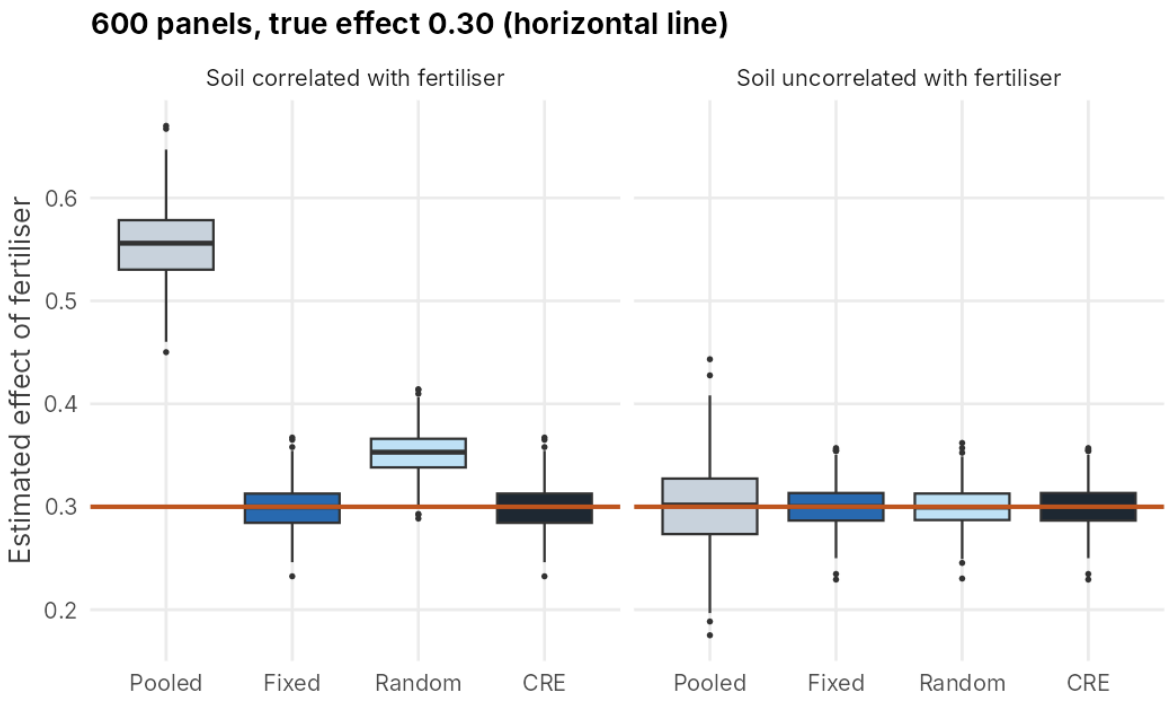
```
four_estimates <- function(d) {
  e <- fe_re(d, "farm", "lyield", "lfert")
  xb <- ave(d$lfert, d$farm)
  c(Pooled = unname(coef(lm(lyield ~ lfert, data = d))["lfert"]),
    Fixed  = unname(e["b_fe"]),
    Random = unname(e["b_re"]),
    CRE    = unname(coef(lm(lyield ~ lfert + xb, data = d))["lfert"]))
}

set.seed(2026)
run_mc <- function(rho, reps = 600)
  t(replicate(reps, four_estimates(make_panel(rho = rho))))

mc <- rbind(
  data.frame(run_mc(0.6), world = "Soil correlated with fertiliser"),
  data.frame(run_mc(0.0), world = "Soil uncorrelated with fertiliser"))

methods <- c("Pooled", "Fixed", "Random", "CRE")
long <- do.call(rbind, lapply(methods, function(m)
  data.frame(world = mc$world, method = m, estimate = mc[[m]])))
long$method <- factor(long$method, levels = methods)

ggplot(long, aes(method, estimate, fill = method)) +
  geom_boxplot(show.legend = FALSE, outlier.size = 0.6) +
  geom_hline(yintercept = 0.30, colour = book_palette$reject, linewidth = 0.9) +
  scale_fill_manual(values = c(book_palette$soft, book_palette$accent,
                               book_palette$fill, book_palette$ink)) +
  facet_wrap(~ world) +
  labs(x = NULL, y = "Estimated effect of fertiliser",
       title = "600 panels, true effect 0.30 (horizontal line)") +
  theme_book()
```



plot of chunk pd-mc

```
summ <- function(d) round(rbind(mean = colMeans(d[, 1:4]),
                                sd   = apply(d[, 1:4], 2, sd)), 3)
list(correlated   = summ(mc[mc$world == "Soil correlated with fertiliser", ]),
     uncorrelated = summ(mc[mc$world == "Soil uncorrelated with fertiliser", ]))
```

```
#> $correlated
#>      Pooled Fixed Random   CRE
#> mean  0.555 0.299  0.353 0.299
#> sd    0.037 0.021  0.021 0.021
#>
#> $uncorrelated
#>      Pooled Fixed Random  CRE
#> mean  0.301  0.30  0.300 0.30
#> sd    0.039  0.02  0.019 0.02
```

The pattern is the whole content of this unit.

- **Soil correlated with fertiliser.** Pooled and random effects are biased; fixed effects and CRE are centred on 0.30.
- **Soil uncorrelated with fertiliser.** All four are unbiased. Random effects is the most precise; fixed effects and CRE give up a little precision by ignoring the between variation, and pooled OLS is the *least* precise of all, because it ignores the serial correlation.

CRE gets the consistency of fixed effects for the time-varying coefficient. What it does not do is make fixed effects any *more efficient* — its slope has the same spread. Its extra value is the test of $\psi = 0$ and the ability to include time-constant variables.

Real data: profit and scale in Indian states

> *Do industrial profits rise with the size of a state's manufacturing sector?*

The file `asi-state-profit.csv` has the gross value added (GVA) and profit of registered manufacturing in each state, by year, taken from the Annual Survey of Industries. We ask whether, in a year when a state's manufacturing sector is larger, a larger fraction of its value added is retained as profit.

The outcome is the **profit share** (profit divided by GVA), the explanatory variable the log of GVA. We keep the 21 states observed in all 43 years from 1980 to 2023 so that the panel is balanced.

```
asi <- read.csv("data/asi-state-profit.csv")
years_seen <- table(asi$state)
asi <- asi[asi$state %in% names(years_seen)[years_seen == 43], ]
asi$lgva <- log(asi$gva)
asi$xbar <- ave(asi$lgva, asi$state)

c(states = length(unique(asi$state)), rows = nrow(asi))
```

```
#> states   rows
#>     21    903
```

Profit share is bounded above by 1 and can be strongly negative in bad years, so a few extreme values will matter.

```
pooled_a <- lm(profit_share ~ lgva, data = asi)
fe_a     <- lm(profit_share ~ lgva + factor(state), data = asi)
cre_a    <- lm(profit_share ~ lgva + xbar, data = asi)
est_a    <- fe_re(asi, "state", "profit_share", "lgva")

res <- rbind(
  Pooled = c(coef(pooled_a)["lgva"], cluster_se(pooled_a, asi$state)["lgva"]),
  Fixed  = c(coef(fe_a)["lgva"],     cluster_se(fe_a, asi$state)["lgva"]),
  Random = c(est_a["b_re"], NA),
  CRE    = c(coef(cre_a)["lgva"],    cluster_se(cre_a, asi$state)["lgva"]))
colnames(res) <- c("Estimate", "Clustered SE")
round(res, 4)
```

```
#>        Estimate Clustered SE
#> Pooled   0.0618       0.0192
#> Fixed    0.0826       0.0114
#> Random   0.0803           NA
#> CRE      0.0826       0.0114
```

The pooled slope is 0.062; fixed effects is 0.083. A state where manufacturing is larger than *its own* usual level retains about 0.8 percentage points more of value added as profit for each 10% rise in GVA (0.083×0.10). Pooled OLS understates it, since it also compares states that differ in size for other reasons.

Now test whether the random effects assumption holds.

```
        round(hausman(est_a), 4)
```

```
#>       H p_value
#>  3.5031  0.0613
```

```
        se_psi <- cluster_se(cre_a, asi$state)["xbar"]
        round(c(psi_hat = unname(coef(cre_a)["xbar"]), clustered_se = unname(se_psi),
                t = unname(coef(cre_a)["xbar"] / se_psi)), 3)
```

```
#>      psi_hat clustered_se            t
#>       -0.032        0.027       -1.195
```

The Hausman test is borderline (p close to 0.06) and the clustered test on $\hat{\psi}$ does not reject. Different questions, different answers, and here neither is decisive. With only 21 states, no test is going to be definitive.

Time matters too

Everything so far removed what is constant over time within a state. But profit shares, GVA and prices all drift over 43 years for reasons that affect every state at once: inflation, liberalisation, a changing mix of industries. Anything common to all states in a year can be absorbed by adding a **dummy for each year** — the unit-and-time version of fixed effects, called **two-way fixed effects**.

```
        fe2_a  <- lm(profit_share ~ lgva + factor(state) + factor(yr), data = asi)
        cre2_a <- lm(profit_share ~ lgva + xbar + factor(yr), data = asi)

        round(rbind(
          `Fixed, state only`    = c(coef(fe_a)["lgva"],  cluster_se(fe_a, asi$state)
["lgva"]),
          `Fixed, state + year`  = c(coef(fe2_a)["lgva"], cluster_se(fe2_a, asi$state)
["lgva"]),
          `CRE: coefficient on x` = c(coef(cre2_a)["lgva"], cluster_se(cre2_a, asi$state)
["lgva"]),
          `CRE: coefficient on mean` = c(coef(cre2_a)["xbar"], cluster_se(cre2_a, asi$state)
["xbar"])),
          4)
```

```
#>                            lgva   lgva
#> Fixed, state only         0.0826 0.0114
#> Fixed, state + year       0.1824 0.0294
#> CRE: coefficient on x     0.1824 0.0294
#> CRE: coefficient on mean -0.1319 0.0555
```

The estimate **more than doubles**, to 0.182, once year effects are included. GVA is measured in current rupees, so it rises in every state every year with inflation, and the profit share has its own trend; a regression that ignores common time effects mixes that trend into the slope. After removing it, a state's own deviation from its yearly norm is what

remains. And now the CRE coefficient on the mean is different from zero ($t \approx -2.4$), so the random effects assumption fails once time is controlled for.

WATCH OUT

The two sets of results are a reminder that a specification choice can change the answer as much as the choice of estimator. Neither state nor year dummies will remove a **time-varying, state-specific** influence — a state's policy change in 1995, a new industrial estate — and a causal reading of 0.18 would need an argument that no such influence moves with GVA. Fixed effects supports the claim "within a state, in years when GVA is high, the profit share is high". It does not by itself support "raising GVA raises profits".

Choosing and using the estimators

Question	If yes	If no
Could a_i be correlated with the regressors?	Fixed effects or CRE	Random effects is possible
Do you need the effect of a time-constant variable?	CRE (with an assumption)	Fixed effects
Is there a trend or shock common to all units?	Add time dummies	—
Is T very small and N large?	Cluster on the unit	Take extra care with inference

Some practical principles.

1. **Default to fixed effects** (or CRE) when the question is causal. The random effects assumption — that the unobserved effect is unrelated to everything in the model — is rarely credible for the questions economists ask.
2. **Report the Mundlak test, not only the Hausman test**, because it survives clustering.
3. **Fixed effects removes only what is constant over time.** An omitted variable that changes within the unit, in a way related to x, still biases the estimate.
4. **Cluster standard errors on the unit.** Always.
5. **Check how much within variation there is.** If x barely changes within units, the fixed effects estimate will be imprecise, and measurement error will be a larger share of what does vary.
6. **Do not interpret the unit dummies.** They are controls, not findings.

GOING DEEPER

When the regressors include a **lagged dependent variable**, $y_{i,t-1}$, strict exogeneity fails by construction, and fixed effects is biased. The bias shrinks at rate $1/T$ (the "Nickell bias"), so it is small in a long panel and large in a short one. The usual remedies are instrumental-variable estimators in differences. They go beyond this book, but you should know to be suspicious of any short-panel regression with a lagged outcome on the right-hand side.

Summary

KEY IDEA

- **Panel data** follow the same units over several periods. A unit can then be compared with itself, which removes everything about it that stays constant.
- The model is $y_{it} = \beta_0 + \beta_1 x_{it} + a_i + u_{it}$. a_i is the **unobserved effect**, constant over time; u_{it} is the **idiosyncratic error**; their sum is the **composite error**.
- **Pooled OLS** is biased when a_i is correlated with x, and its standard errors are wrong because the composite errors of one unit are correlated. Cluster-robust standard errors fix only the second problem.
- The **fixed effects (within) estimator** subtracts each unit's time average from every variable, so a_i drops out, and runs OLS. It permits arbitrary correlation between a_i and x. Degrees of freedom are $N(T-1)-k$.
- It is **identical** to OLS with a dummy for every unit, and with $T = 2$ it is identical to first differencing.
- It uses **only within variation**, so it cannot estimate the effect of any variable constant over time, and it can worsen measurement error.
- A unit observed once contributes nothing. **Attrition** related to a_i is harmless; attrition related to u_{it} is not.
- **Random effects** assumes $\mathrm{Cov}(x_{it}, a_i) = 0$ and estimates by GLS, subtracting a fraction θ of each unit's mean. It lies between pooled OLS ($\theta = 0$) and fixed effects ($\theta = 1$), and is the most efficient estimator **only if** its assumption holds.
- The **Hausman test** compares fixed and random effects. It is invalid under clustering and heteroskedasticity.
- **Correlated random effects** adds the unit means of x to the regression. The coefficient on x equals the fixed effects estimate, the coefficient on the mean tests the random effects assumption robustly, and time-constant variables become estimable — under a stronger assumption about them.
- In the simulation, pooled and random effects were biased (about 0.55 and 0.35 against 0.30) when soil was related to fertiliser; fixed effects and CRE were not.

Practice problems

YOUR TURN

Work each one before opening the solution. The question bank that follows has no answers.

Numerical problems

11.1 Three farms are each observed in three seasons. x is fertiliser and y is yield.

Farm	x_{i1}	x_{i2}	x_{i3}	y_{i1}	y_{i2}	y_{i3}
1	1	2	3	2	3	7
2	4	5	6	7	10	10
3	2	4	6	5	9	10

i. Compute each farm's mean of x and of y.
ii. Compute the demeaned values $\ddot{x}_{it}$ and $\ddot{y}_{it}$.
iii. Compute the fixed effects estimate $\hat{\beta}_1$.
iv. Is the pooled OLS slope larger or smaller than your answer? Why?

ANSWER

(i) $\bar{x} = 2, 5, 4$ and $\bar{y} = 4, 9, 8$ for farms 1, 2, 3.

(ii) Farm 1: $\ddot{x} = (-1, 0, 1)$, $\ddot{y} = (-2, -1, 3)$. Farm 2: $\ddot{x} = (-1, 0, 1)$, $\ddot{y} = (-2, 1, 1)$. Farm 3: $\ddot{x} = (-2, 0, 2)$, $\ddot{y} = (-3, 1, 2)$.

(iii) $\sum \ddot{x}\ddot{y} = (2 + 0 + 3) + (2 + 0 + 1) + (6 + 0 + 4) = 18$ and $\sum \ddot{x}^2 = 2 + 2 + 8 = 12$, so $\hat{\beta}_1^{FE} = 18/12 = 1.5$.

(iv) The pooled slope is 1.615, larger than 1.5. Farms with higher x also have higher average y, in a way the within-farm movements do not account for: farm 2 has both the largest x and the largest y on average. Pooled OLS credits all of that to x, which is the same composite-error bias as in the simulation. Check: `lm(y ~ x + factor(f))` reproduces 1.5.

11.2 A panel has $N = 50$ units and $T = 6$ periods. A fixed effects model is estimated by running OLS on demeaned data with three regressors, and the software reports a standard error of 0.040 for one coefficient.

i. How many degrees of freedom did the regression use, and how many should it have used?
ii. By what factor should the reported standard error be multiplied?
iii. Give the corrected standard error.

ANSWER

(i) It used $NT - k = 300 - 3 = 297$. The correct figure is $NT - N - k = 300 - 50 - 3 = 247$.

(ii) The estimate of σ_u^2 is a sum of squared residuals divided by the degrees of freedom, so the standard error scales with $\sqrt{297/247} = 1.0966$.

(iii) $0.040 \times 1.0966 = 0.0439$.

11.3 A researcher estimates a log-wage equation on a panel of workers with fixed effects. For each variable, say whether its coefficient can be estimated, and why.

i. Union membership, which changes for some workers.
ii. Gender.

iii. Years of schooling, in a panel of adults whose schooling is complete.
iv. Years of work experience, with a full set of year dummies, where every worker is employed in every year.
v. Years of schooling interacted with the year dummies.

ANSWER

(i) Yes: it varies within workers.

(ii) No. It is constant for each worker, so it is removed by the within transformation.

(iii) No, for the same reason: constant for each adult.

(iv) No. Experience rises by one for every worker every year, so after the worker's effect is removed it is perfectly collinear with the year dummies.

(v) Yes, but only the *differences* in the return to schooling between each year and the base year. The return in any single year is not identified.

11.4 Estimated variance components in a random effects model are $\hat{\sigma}_a^2 = 0.04$ and $\hat{\sigma}_u^2 = 0.16$.

i. Compute θ for $T = 4$.
ii. Compute θ for $T = 16$.
iii. What does the difference tell you about how random effects relates to fixed effects as T grows?
iv. Compute the correlation between two composite errors from the same unit.

ANSWER

(i) $\theta = 1 - \sqrt{0.16/(0.16 + 4 \times 0.04)} = 1 - \sqrt{0.5} = 0.293$.

(ii) $\theta = 1 - \sqrt{0.16/(0.16 + 16 \times 0.04)} = 1 - \sqrt{0.2} = 0.553$.

(iii) θ rises with T: random effects subtracts more of each unit's mean and moves towards fixed effects ($\theta = 1$).

(iv) $0.04/(0.04 + 0.16) = 0.2$.

11.5 Fixed and random effects estimates of one slope are $\hat{\beta}^{FE} = 0.30$ with standard error 0.040, and $\hat{\beta}^{RE} = 0.35$ with standard error 0.030.

i. Compute the Hausman statistic.
ii. The 5% critical value of χ^2_1 is 3.84. What do you conclude?
iii. Why would you not be reassured if a colleague reported the result of this test from a regression with clustered standard errors?

ANSWER

(i) $H = (0.30 - 0.35)^2 / (0.040^2 - 0.030^2) = 0.0025/0.0007 = 3.571$.

(ii) It is below 3.84, so we do not reject the random effects assumption at 5% ($p \approx 0.059$). It is close to the boundary, so the conclusion is fragile.

(iii) The formula takes the difference of the two variances, which is justified only if random effects is efficient. With clustered standard errors that is no longer true, and the denominator can be negative or meaningless. The regression form of the test, adding the unit means, avoids the problem.

11.6 A CRE regression of log wage on union status (time-varying) and its unit mean gives the following, with clustered standard errors.

	Estimate	SE
union	0.08	0.02
mean of union	0.14	0.05

i. What is the fixed effects estimate of the union effect?
ii. What is the between effect of union status?
iii. Test whether random effects is valid, and state the conclusion.

ANSWER

(i) 0.08. In a CRE regression the coefficient on the time-varying variable equals the fixed effects estimate.

(ii) The coefficient on the mean is the between effect minus the within effect, so the between effect is $0.14 + 0.08 = 0.22$.

(iii) $t = 0.14/0.05 = 2.8$, which exceeds 1.96. Reject $\psi = 0$: the unobserved effect is related to the unit's average union status, and random effects would be biased. Workers who are more often unionised earn more than the within-worker change in union status can explain, probably because of unobserved differences between workers.

11.7 In a pooled regression, $\mathrm{Var}(x) = 0.40$ and $\mathrm{Cov}(x, a) = 0.12$. The true $\beta_1 = 0.50$.

i. What does pooled OLS estimate in a large sample?
ii. What would it estimate if $\mathrm{Cov}(x, a) = 0$?
iii. What does fixed effects estimate in the first case?

ANSWER

(i) $0.50 + 0.12/0.40 = 0.80$.

(ii) 0.50 — the unobserved effect no longer biases it, though the standard errors are still wrong.

(iii) 0.50, whatever the covariance. Fixed effects allows any correlation between a_i and x.

11.8 Four units are each observed in two periods.

Unit	x_1	x_2	y_1	y_2
1	1	2	3	6
2	2	2	4	5
3	3	5	7	11
4	4	6	8	12

i. Compute the first-difference slope, with an intercept.
ii. Verify that the fixed effects estimate is the same.

ANSWER

(i) $\Delta x = (1, 0, 2, 2)$, $\Delta y = (3, 1, 4, 4)$, with means 1.25 and 3. Then $\sum(\Delta x - 1.25)(\Delta y - 3) = 4.0$ and $\sum(\Delta x - 1.25)^2 = 2.75$, so the slope is $4.0/2.75 = 1.4545$.

(ii) Fixed effects with a dummy for the second period, as the first difference regression with an intercept requires, gives 1.4545. The two are identical when $T = 2$.

R exercises

11.9 Using the simulation function of the section on what panel data look like, set `rho = 0` and `N = 100`, `T = 3`. Compute the pooled, fixed effects, random effects and CRE estimates and the Hausman statistic. Does the Hausman test reject? Should it?

ANSWER

```
d <- make_panel(N = 100, T = 3, rho = 0, seed = 5)
e <- fe_re(d, "farm", "lyield", "lfert")
round(e, 4)
hausman(e)

xb <- ave(d$lfert, d$farm)
coef(lm(lyield ~ lfert + xb, data = d))
```

With `rho = 0` the random effects assumption is true, so the test should not reject, and 5% of the time it will anyway — that is what a 5% significance level means. Run it with several seeds and count the rejections.

11.10 Using `asi-state-profit.csv`, estimate the fixed effects regression of profit share on log GVA using **all** states (the unbalanced panel), with clustered standard errors. How many states contribute? Compare with the balanced panel in the text.

ANSWER

```
asi_all <- read.csv("data/asi-state-profit.csv")
asi_all$lgva <- log(asi_all$gva)

table(table(asi_all$state))                # how many years each state appears

fe_all <- lm(profit_share ~ lgva + factor(state), data = asi_all)
c(coef(fe_all)["lgva"], cluster_se(fe_all, asi_all$state)["lgva"])
```

The table shows that states observed in only one year contribute nothing, since their demeaned values are zero. Check which dropped out by counting states with a single row. The estimate differs from the balanced-panel one because the extra states have fewer years and a different mix of industries — and, if their absence from the full panel is related to outcomes, because of the selection problems discussed under unbalanced panels.

Question bank

WATCH OUT

No answers are given for this section, by design.

Multiple choice

QB 11.1 In the model $y_{it} = \beta_1 x_{it} + a_i + u_{it}$, the term a_i is

a. the idiosyncratic error
b. constant over time for unit i
c. observed for each i
d. uncorrelated with x_{it} by definition

QB 11.2 The within transformation

a. subtracts the overall mean from each observation
b. subtracts each unit's time average from its observations
c. subtracts the previous period's value from each observation
d. adds a dummy for each time period

QB 11.3 The fixed effects estimator requires which of the following? *(select all that apply)*

a. a_i uncorrelated with x_{it}
b. u_{it} uncorrelated with x_{is} for all periods s
c. variation in x_{it} over time within units
d. a balanced panel

QB 11.4 In a fixed effects regression with N units, T periods and k time-varying regressors, the degrees of freedom are

a. $NT - k$

b. $NT - N - k$
c. $N(T - 1)$
d. $NT - T - k$

QB 11.5 Which of the following can be estimated by a fixed effects regression? *(select all that apply)*

a. the effect of gender in a panel of workers
b. the effect of a state's rainfall in each year
c. the effect of years of schooling in a panel of adults
d. the effect of union membership for workers who change status

QB 11.6 With two periods, the first difference estimator with an intercept is

a. more efficient than fixed effects
b. less efficient than fixed effects
c. identical to fixed effects with a dummy for the second period
d. biased if a_i is correlated with x

QB 11.7 In the random effects estimator, $\theta = 0$ corresponds to

a. fixed effects
b. pooled OLS
c. the between estimator
d. first differences

QB 11.8 The coefficient on $\bar{x}_i$ in a Mundlak regression equals

a. the between effect
b. the within effect
c. the between effect minus the within effect
d. the fixed effect for unit i

Short reasoning

QB 11.9 Explain, without algebra, why the fixed effects estimator is not biased by a characteristic of the unit that stays constant and is unrelated to anything observed.

QB 11.10 A researcher says: "I include a dummy for every district, so I have controlled for rainfall in each district's history." Is this fully correct? What does it control for and what not?

QB 11.11 Explain why a unit observed in only one period plays no role in a fixed effects estimation, even though it is in the data.

QB 11.12 Why can fixed effects be *more* damaged by measurement error in x than pooled OLS?

QB 11.13 A fixed effects regression is estimated with a full set of year dummies. A colleague asks for the effect of age in a panel in which everyone is observed annually. Explain why this cannot be estimated.

QB 11.14 State the assumption on which random effects relies and give one reason it might fail in a panel of households.

QB 11.15 Describe two advantages of the Mundlak regression test over the Hausman test.

QB 11.16 A panel of firms loses firms that go bankrupt. Under what circumstances does this attrition bias a fixed effects estimate, and under what circumstances does it not?

Numerical

QB 11.17 Two farms are observed over three seasons. Farm A has $x = (2, 4, 6)$ and $y = (5, 9, 10)$; farm B has $x = (1, 2, 3)$ and $y = (2, 3, 7)$. Compute the fixed effects slope and the pooled OLS slope.

QB 11.18 A fixed effects regression on 120 households observed for 4 years gives a residual sum of squares of 54 from the demeaned regression with 2 regressors. Compute $\hat{\sigma}_u^2$ with the correct degrees of freedom.

QB 11.19 With $\hat{\sigma}_a^2 = 0.36$ and $\hat{\sigma}_u^2 = 0.09$, compute θ for $T = 5$ and the correlation between two composite errors from the same unit.

QB 11.20 A CRE regression of log output on log capital and its unit mean gives coefficients 0.35 (SE 0.06) and 0.20 (SE 0.07), clustered. Report the fixed effects estimate, the implied between estimate, and your conclusion about the random effects assumption.

Looking ahead

Panel data buy one thing: protection against everything about a unit that stays constant. They do not protect against what changes.

The next question follows naturally. Suppose a policy is introduced in some states in one year and not in others. A panel of states, observed before and after, with state and year dummies — a **two-way fixed effects** regression — compares how outcomes changed in treated states with how they changed in untreated ones. That comparison of changes is the idea behind **difference-in-differences**, and it is the most widely used application of the methods in this unit.

It rests on an assumption fixed effects alone cannot test: that the treated states, absent the policy, would have followed the same trend as the others. When that is plausible, panel data come closest to the controlled experiment that Unit 6 wished for.